

FIN ST1092–ST1181

Operator-to-Total-State Nonuniqueness and the A -Only No-Go

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This report tests whether the strict operator itself can select the state space and complete dynamics of the one total nadsoliton. It proves a terminal no-go for that A -only route. The no-go does not exclude a richer FIN construction containing independently derived algebraic, nonlinear, refinement-limit or operational data.

Abstract

Release 10.95 showed conditionally that a complete invariant normalized state of the whole nadsoliton cannot reach a nonexistence state outside its state space. The unresolved question was whether the frozen strict operator A itself canonically determines that total-state space and dynamics.

ST1092–ST1181 answer this question negatively for the A -only route. First, the positive result is strengthened: $Q = -A$ satisfies the exact finite Markov generator criterion. All off-diagonal rates are positive, row sums vanish, and e^{-tA} is a positive mass-preserving irreducible reversible Markov semigroup for every $t \geq 0$. Its unique stationary distribution is uniform and it has nonnegative Shannon entropy production.

The main theorem is an operator-to-state nonuniqueness no-go. The same A canonically supports at least three pairwise inequivalent normalized models: the classical simplex Δ_{11} with heat flow, pure quantum rays $\mathbb{CP}^{11}$ with unitary flow, and density operators $\mathcal{D}_{12}$ with unitary conjugation. Their real dimensions are respectively 11, 22 and 143; their boundary/convex structures and stationary sets differ, and the heat and unitary dynamics are inequivalent by invertibility and attractor structure. The inherited $\mathbb{Z}_{12}$ symmetry acts on every candidate and therefore does not select one. Exact $12 \rightarrow 24$ refinements repeat the ambiguity and introduce a free fiber rate.

The heat interpretation also has a precise microscopic boundary. A fixed-time CPTP measure-and-prepare dilation exists, but no single finite closed environment with a time-independent Hamiltonian and fixed initial state can realize the convergent strict heat semigroup exactly for all times: finite closed unitary dynamics is almost periodic. Hence an infinite internal bath, fresh ancillas or a limit procedure is additional structure. Strict heat provides detailed-balance gradient currents, not a sustained internal pump.

Consequently, what blocks annihilation remains model-relative: mass in the classical model, norm/trace in the quantum models, and state-space closure in the ontological theorem. The operator does not decide which blocker is physical. At least two extra axiom classes are unavoidable: a state/observable category and a physical channel with clock/instrument semantics. Repeating A -only state searches is now closed; progress requires genuinely new strict data or an explicitly labelled axiom.

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1 Executive result

The six rounds establish both a positive theorem and a terminal obstruction.

Positive theorem. The strict matrix is not merely mass preserving on sampled vectors. It is a weighted complete-graph Laplacian. Therefore $-A$ is a genuine conservative continuous-time Markov generator and e^{-tA} is stochastic for every $t \geq 0$.

Terminal A -only no-go. The operator datum does not choose whether its normalized states are classical probabilities, quantum rays, or quantum density matrices, nor whether the physical channel is heat, unitary evolution or one of infinitely many trace-preserving dephasing laws. Symmetry and refinement do not remove this ambiguity.

The result terminates the route

$$A \text{ alone} \mapsto (\mathcal{M}_{\text{tot}}, \Phi, \text{operational meaning})$$

as a unique derivation. It does not terminate FIN if an additional object can be independently derived.

2 Input and status discipline

The input is the frozen strict operator

$$A = sI - W, \quad s = 1.660307278766099,$$

with the certified quadratic form

$$\langle f, Af \rangle = \frac{1}{2} \sum_{x,y} W_{xy} |f_x - f_y|^2 \geq 0.$$

This cycle uses neither a silent legacy substitution nor legacy physical-role transfer. The ontology remains one nadsoliton as the whole fundamental being, with light, matter and observer downstream/internal. No selector or QW-2191 closure is imported.

Statuses are used as follows: [Proven] for theorems within declared finite models, [Constructed] for explicit optional structures, [Conditional] when a state category or operational premise is added, and [Open] for absent FIN derivations.

3 Round I: ST1092–ST1106 — state candidates

3.1 Exact Markov certificate

Let $Q = -A$. The numerical reconstruction gives

$$\min_{i \neq j} Q_{ij} = 0.011070817321442113 > 0, \quad \max_{i \neq j} Q_{ij} = 0.4699856726450201,$$

and

$$\|Q\mathbf{1}\|_{\infty} \leq 2.22 \times 10^{-16}.$$

The displayed residual is floating reconstruction error; the defining row-sum identity is exact in the model. Since Q is Metzler and conservative, $P_t = e^{tQ} = e^{-tA}$ is entrywise nonnegative and stochastic. Because every off-diagonal rate is positive, the chain is irreducible. Symmetry gives detailed balance with the uniform distribution $u = \mathbf{1}/12$.

The Dirichlet form was independently replayed:

$$f^T A f = \frac{1}{2} \sum_{i,j} W_{ij} (f_i - f_j)^2.$$

3.2 Three canonical candidates

After the corresponding state category is chosen, the same A gives:

$$\begin{array}{lll} \text{classical:} & \Delta_{11} = \{p \geq 0 : \mathbf{1}^T p = 1\}, & p(t) = e^{-tA} p(0), \\ \text{pure quantum:} & \mathbb{CP}^{11}, & [\psi(t)] = [e^{-itA} \psi(0)], \\ \text{mixed quantum:} & \mathcal{D}_{12} = \{\rho \geq 0 : \text{Tr } \rho = 1\}, & \rho(t) = e^{-itA} \rho(0) e^{itA}. \end{array}$$

Their real dimensions are

$$\dim \Delta_{11} = 11, \quad \dim_{\mathbb{R}} \mathbb{CP}^{11} = 22, \quad \dim_{\mathbb{R}} \mathcal{D}_{12} = 12^2 - 1 = 143.$$

The simplex has faces and corners; projective space is a compact boundaryless manifold; the density-state body is convex with a rank-stratified boundary. They are not homeomorphic state spaces.

The strict spectrum has multiplicities

$$1, 2, 2, 2, 2, 2, 1.$$

Thus the classical heat chain has one stationary probability state, while the Hermitian quantum commutant has real dimension

$$1^2 + 5 \cdot 2^2 + 1^2 = 22,$$

and its trace-one stationary slice has affine dimension 21. The stationary content is therefore also inequivalent.

4 The operator-to-state nonuniqueness theorem

Theorem 1 (A -only state-selection no-go). *Let A be the frozen 12×12 real symmetric strict Laplacian. The datum consisting of A , its inherited vertex basis, its symmetry group and its functional calculus does not determine a unique normalized total-state space and physical channel. In particular, the three constructions (Δ_{11}, e^{-tA}) , $(\mathbb{CP}^{11}, e^{-itA})$ and $(\mathcal{D}_{12}, \rho \mapsto e^{-itA} \rho e^{itA})$ are valid, pairwise inequivalent realisations of the same operator datum.*

Proof. The Markov criterion proves that the first construction is a normalized positive semigroup. Self-adjointness proves that the second and third are a unitary ray flow and a trace-preserving positive conjugation group. The state spaces are pairwise nonhomeomorphic by their distinct covering dimensions and boundary/convex structures. The heat dynamics has a unique global stationary probability attractor and its inverse is not a stochastic channel, whereas the unitary dynamics is a bijective recurrent group and has a higher-dimensional stationary commutant. Hence no state-space/dynamical equivalence identifies the constructions. All inherit the cyclic permutation action, so symmetry does not remove the alternatives. Therefore A and the declared associated data do not select a unique one. $\square$

Corollary 2. *Any unique operational FIN theory requires at least one independently justified state/observable-category premise and one physical-channel/instrument premise, unless richer strict data proves an equivalent selection theorem.*

The theorem is not a claim that the three candidates exhaust all models. Additional candidates strengthen rather than weaken nonuniqueness.

5 Round II: ST1107–ST1121 — completeness and killing

Finite matrix exponentials exist globally. The declared unitary and heat channels therefore have no finite-time blow-up. For linear channels, no-killing is a generator condition:

$$Q\mathbf{1} = 0 \quad (\text{classical}), \quad \mathcal{L}^*(I) = 0 \quad (\text{quantum}).$$

It is not a consequence of the word “self-reference.”

The same A supports infinitely many additional complete quantum channels, for example

$$\mathcal{L}_\gamma(\rho) = -i[A, \rho] + \gamma \left(A\rho A - \frac{1}{2}\{A^2, \rho\} \right), \quad \gamma \geq 0.$$

Every γ is trace preserving, but A does not select γ . Uniform killing supplies the contrasting classical family

$$Q_\gamma = -A - \gamma I, \quad S(t) = e^{-\gamma t}.$$

Again, γ is new input.

The full nonlinear nadsoliton law remains absent. Compactness of a candidate state space is insufficient unless the vector field is tangent/inward, regular and globally solvable. Therefore no universal no-killing theorem for future nonlinear or open extensions follows from the linear certificate.

6 Round III: ST1122–ST1136 — internal heat and dilation

For a classical probability state define the edge current into i from j :

$$J_{ij} = W_{ij}(p_j - p_i) = -J_{ji}.$$

It obeys the exact continuity equation

$$\dot{p}_i = \sum_j J_{ij} = -(Ap)_i, \quad \sum_i \dot{p}_i = 0.$$

Shannon entropy satisfies

$$\frac{dH}{dt} = \frac{1}{2} \sum_{i,j} W_{ij}(p_i - p_j)(\log p_i - \log p_j) \geq 0.$$

Because the chain is irreducible and detailed-balanced, a stationary state is uniform and every stationary edge current vanishes. Strict heat is therefore a relaxational gradient flow, not a sustained internal pump or circulation.

6.1 Fixed-time quantum embedding

For $P = e^{-A}$, one CPTP embedding is the measure-and-prepare channel with Kraus operators

$$K_{ij} = \sqrt{P_{ij}} |i\rangle\langle j|, \quad \sum_{i,j} K_{ij}^* K_{ij} = I,$$

so

$$\mathcal{E}(\rho) = \text{diag}(P \text{diag } \rho).$$

All 144 entries of P are positive, and this declared channel has full Choi rank 144. It destroys input coherence. Other quantum extensions of the same classical transition matrix exist, so this is not a state-category selector.

Proposition 3 (finite closed dilation obstruction). *No single finite-dimensional environment in a fixed initial state, coupled by one time-independent total Hamiltonian, realizes the nontrivial strict mixing heat semigroup exactly for every $t \geq 0$.*

Proof. For finite system and environment, every matrix element of the reduced dynamics is a finite sum of exponentials of energy differences and is almost periodic. A nonstationary strict heat trajectory converges exponentially to the unique uniform state. A nonconstant almost-periodic function cannot have a one-sided limit. The two behaviours are incompatible. $\square$

A fixed-time Stinespring dilation remains valid. Exact all-time irreversible heat requires an infinite internal bath, fresh internal ancillas, or a continuum/singular limit. Such structure may remain internal to the one total nadsoliton, but it is not supplied by A alone.

7 Round IV: ST1137–ST1151 — refinement and duality

An exact two-point-fiber family is

$$A_{24}(q) = A_{12} \otimes I_2 + I_{12} \otimes \begin{pmatrix} q & -q \\ -q & q \end{pmatrix}, \quad q \geq 0.$$

Let C sum each two-point fiber. Then

$$CA_{24}(q) = A_{12}C,$$

and consequently

$$Ce^{-tA_{24}(q)} = e^{-tA_{12}}C, \quad Ce^{-itA_{24}(q)} = e^{-itA_{12}}C.$$

The dual functional calculi are exactly compatible with this refinement. However, every $q \geq 0$ gives the same coarse dynamics. At $q = 0$ the refined kernel dimension is two; for $q > 0$ it is one. Fine topology/connectivity can therefore change while the coarse observer is exactly blind.

The spectrum is

$$\text{spec } A_{24}(q) = \text{spec } A_{12} \cup (\text{spec } A_{12} + 2q).$$

For a fine classical state, entropy decomposes as

$$H(p_{24}) = H(Cp_{24}) + \sum_i (Cp_{24})_i H(p_{\text{fiber}|i}).$$

This makes hidden information precise. Yet each $A_{24}(q)$ again admits simplex, ray and density-state models. Refinement repeats rather than resolves the state-category ambiguity. The parameter q is only a relative inverse time scale until a dimensional anchor is supplied.

8 Round V: ST1152–ST1166 — observer, clock and record

A single perfectly distinguishable 12-state carrier stores at most

$$\log_2 12 = 3.5849625007 \dots$$

classical bits. The Holevo bound gives the same single-use upper bound for classical information extracted from a 12-dimensional quantum system. A literal single 12-level model cannot contain an unbounded independent record history. A complete record-bearing model therefore needs composition, repeated carriers, unbounded refinement or an infinite algebra. This does not exclude such a FIN completion; it excludes identifying the lone finite A_{12} model with the whole operational universe.

Two natural measurements illustrate further ambiguity. The vertex POVM $E_x = |x\rangle\langle x|$ has twelve outcomes but is not informationally complete for density matrices. The spectral PVM has seven effects because of degeneracies and does not resolve vertices. Even a fixed POVM does not fix an instrument:

$$\mathcal{I}_x^L(\rho) = E_x \rho E_x, \quad \mathcal{I}_x^{\text{MP}}(\rho) = \text{Tr}(E_x \rho) \sigma_x$$

have the same outcome probabilities and different postmeasurement states. A does not select the states σ_x , apparatus, ancilla or record.

The spectral gap gives two dimensionless clock candidates:

$$\text{unitary phase: } e^{-it\lambda_1}, \quad T_1 = \frac{2\pi}{\lambda_1} = 8.331 \dots,$$

$$\text{heat contrast: } a(t) = a_0 e^{-\lambda_1 t}.$$

The first is periodic and phase-valued; the second is monotone but saturates and is irreversible. They share a number, not an operational clock. Neither produces SI seconds.

An internal observer also requires a subalgebra or factorization, memory states, an interaction/instrument and a record channel. There is no canonical system–observer split in A_{12} .

9 Round VI: ST1167–ST1181 — terminal verdict

The complete falsification matrix is:

Proposed selector	Test	Result
positivity/mass	strict Markov criterion	selects a valid classical model, not uniqueness
self-adjointness/norm	unitary theorem	selects valid quantum models, not uniqueness
$\mathbb{Z}_{12}$ symmetry	action on candidates	shared by all candidates
spectral stationary structure	fixed-set comparison	candidates remain inequivalent
dual dynamics	heat/unitary calculus	supplies two incompatible channels
exact refinement	$12 \rightarrow 24$ intertwining	ambiguity repeats; q remains free
measurement	POVM/instrument audit	apparatus and poststate remain free
closed internal dilation	recurrence test	finite all-time heat dilation impossible

Thus the current A -only lane has reached a total no-go. This is the requested terminal scientific outcome for that lane, not a rhetorical stop.

10 Consequences for nadsoliton annihilation

Release 10.95 remains valid at its conditional level. Once a state model is chosen:

Model	Blocker	What may still disappear
classical heat on Δ_{11}	conserved mass	contrast, localization, records
pure quantum rays	conserved norm / absence of zero ray	phases relative to coarse access, local patterns
density operators	conserved trace/positivity	coherence under added dephasing, subsystem purity
ontological total-state model	† excluded and flow complete/invariant	internal sectors and accessible structure

The new result adds a decisive qualification:

A proves blockers inside several models but does not select which model is physical.

Therefore the phrase “what blocks annihilation?” has no unique physical answer from A alone. It has several exact mathematical answers and one conditional ontological answer.

11 Minimal unavoidable extra axioms

At least two axiom classes are necessary for a unique operational theory:

1. **State/observable category:** classical probability space, complex Hilbert rays, a density-operator algebra, or another explicitly defined category.
2. **Physical channel and operational semantics:** which semigroup or group represents time, together with clock, preparation, instrument and record rules.

Removing the first leaves the three nonhomeomorphic state spaces. Removing the second leaves heat, unitary and infinitely many dephasing/open channels on compatible carriers. Hence both classes are necessary in the declared problem. They need not be arbitrary assumptions if future FIN mathematics derives them from genuinely new strict data.

12 Research after the no-go

Further local work is admissible only if it leaves the exhausted A -only class. Recommended programmes are:

1. derive an observable algebra from independently defined coupling composition rather than functional calculus of A ;
2. test whether the adaptive law supplies a state-category selector;
3. construct and certify a full nonlinear invariant total-state manifold;
4. derive an infinite internal dilation from a refinement limit;
5. seek a unique refinement-limit state functor with an explicit universality property;
6. formulate classical-versus-quantum operational discriminants;
7. construct observer, apparatus and record algebras inside the total state;
8. derive a relational clock and dimensional anchor together;
9. audit full strict-kernel data beyond A for categorical information;
10. formalize the no-go theorem in a proof assistant;
11. stop any candidate that reduces to another function of A ;
12. postpone laboratory transfer until one operational model is uniquely specified.

13 Final verdict

The strict operator is richer than a generic symmetric matrix: it is simultaneously a valid graph/Markov Laplacian and a self-adjoint quantum generator. That dual success is exactly why it cannot select the ontology. The same datum generates multiple exact, inequivalent normalized worlds. No amount of repeating its spectrum, symmetry, functional calculus or the tested refinement removes the ambiguity. The A -only bridge to a unique total-state theory is therefore mathematically impossible. A state category and an operational dynamics must be independently derived or explicitly postulated.

This is a total no-go for the declared current-datum lane and an important structural result for FIN. It is not a no-go for every possible enriched FIN theory and it is not evidence that any candidate describes nature.

A Six-round ledger

Range	Main object	Verdict
ST1092–ST1106	three canonical state candidates	exact Markov certificate; state nonuniqueness
ST1107–ST1121	complete/no-killing channels	declared linear laws complete; full non-linear law open
ST1122–ST1136	currents and dilation	detailed-balance relaxation, no pump; finite all-time dilation no-go
ST1137–ST1151	two-fiber refinement	exact dual intertwining with free q
ST1152–ST1166	observer, clock and record	finite capacity and instrument ambiguity
ST1167–ST1181	terminal theorem	total no-go for unique state ontology from A alone

B Reproducibility

The six authoritative result sets run from `FIN_ST1092_ST1106_Results.json` through `FIN_ST1167_ST1181_Results.json`. Each contains fifteen programmes and references its individual JSON packet by SHA-256. The combined test `test_fin_st1092_st1181.py` checks all ninety packet hashes, the Markov certificate, nonuniqueness, dilation obstruction, refinement intertwiners, capacity boundary and no-go scope.

C Nonclaims

No laboratory event, external audit, unique physical state category, SI scale, selector, QW-2191 discharge, legacy-to-strict bridge, legacy role transfer, photon, matter law, Standard Model, gravity, L_{total} or Theory of Everything is established. All future FIN PDFs remain English.

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